

AAPD Guideline on Use of Local Anesthesia for Pediatric Dental Patients

Maximum Recommended Dosages (MRD) & Weight-Based Calculation Reference
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1. General Principles

Local anesthetic overdose is one of the most common causes of drug toxicity in pediatric dentistry because children have significantly lower body weight than adults but are sometimes administered similar cartridge volumes. The maximum recommended dose (MRD) must always be calculated using the child's **actual body weight in kilograms**, and the clinician must select the *lower* of the weight-based MRD or the absolute (ceiling) MRD in milligrams.

2. Maximum Recommended Dosages by Agent

Agent / Concentration	mg per cartridge (1.8 mL)	MRD (mg/kg)	Absolute Max (mg)	Age restriction
Lidocaine 2% plain	36	4.4	300	—
Lidocaine 2% + 1:100,000 epinephrine	36	4.4	300	—
Lidocaine 2% + 1:50,000 epinephrine	36	4.4	300	—
Mepivacaine 3% plain	54	4.4	300	—
Mepivacaine 2% + 1:100,000 epinephrine	36	4.4	300	—
Prilocaine 4% plain	72	8.0	400	Avoid in methemoglobinemia/anemia
Articaine 4% + 1:100,000 epinephrine	72	7.0	— (calc. by wt.)	Not recommended < 4 yrs
Bupivacaine 0.5% + 1:200,000 epinephrine	9	1.3	90	Not recommended < 12 yrs

Note: The AAPD monograph lists conservative (lower) MRD values than the FDA package-insert figures used for healthy adults; pediatric protocols should default to the lower, weight-based AAPD values shown above.

3. Worked Example — Lidocaine 2% with 1:100,000 Epinephrine

Patient: 20 kg child (~44 lb).

Step 1 — Weight-based MRD: $20 \text{ kg} \times 4.4 \text{ mg/kg} = 88 \text{ mg}$.

Step 2 — Compare to absolute ceiling: $88 \text{ mg} < 300 \text{ mg}$ absolute max, so 88 mg governs.

Step 3 — Convert to cartridges: $88 \text{ mg} \div 36 \text{ mg/cartridge} = 2.44$ cartridges maximum.

Conclusion: Do not exceed approximately 2½ cartridges of 2% lidocaine with 1:100,000 epinephrine for this patient at a single appointment.

4. Signs of Local Anesthetic Systemic Toxicity (LAST)

- Early / mild: circumoral numbness, metallic taste, tinnitus, lightheadedness, visual disturbance, agitation.
- Progressive CNS: slurred speech, muscle twitching, tremors, generalized tonic-clonic seizure.

- Cardiovascular: initially hypertension/tachycardia, followed by hypotension, bradycardia, arrhythmia, and cardiovascular collapse in severe overdose.
- Management: stop injection, call for help/EMS, support airway/breathing/circulation, administer oxygen, consider IV lipid emulsion (“intralipid”) therapy for severe cardiovascular toxicity per ACLS/PALS-aligned protocol.

5. Additional Precautions in Pediatric Patients

- Aspirate prior to injection to reduce risk of intravascular administration.
- Use a topical anesthetic at the injection site prior to needle insertion.
- Consider mandibular block volume reduction techniques in very young or medically compromised children.
- Educate caregivers on the risk of self-inflicted soft-tissue (lip/cheek/tongue) biting injury while anesthesia is still active, particularly with longer-acting agents such as bupivacaine.
- Document total volume and total milligrams of local anesthetic administered in the patient chart at every visit.

Reference: American Academy of Pediatric Dentistry. Guideline on Use of Local Anesthesia for Pediatric Dental Patients. The Reference Manual of Pediatric Dentistry. Chicago, Ill.: AAPD; latest edition.